Biological Sequence Clustering

Method Description & Technical Reference

Upstream & Downstream Sequence Analysis

Using Pairwise Alignment · UMAP Embedding · K-Means & Hierarchical Clustering

1. Overview

This document describes a computational pipeline for clustering biological sequences extracted from genomic peak regions. The pipeline is designed to process tab-separated (TSV) files containing paired sequence information and produce cluster assignments that reveal shared structural or functional properties among sequences.

The two sequence columns analysed are:

- Upstream – the nucleotide sequence flanking the matched motif on the 5' (upstream) side
- Downstream – the nucleotide sequence flanking the matched motif on the 3' (downstream) side

The pipeline operates in three clustering modes:

1. Based on Upstream sequences only
2. Based on Downstream sequences only
3. Based on both sequences combined (weighted-average distance)

Key design principle: Because multiple GeneIDs can map to the same genomic peak (and therefore share identical Upstream/Downstream sequences), the pipeline first deduplicates on unique sequence pairs, clusters the unique pairs, and then maps cluster labels back to every GeneID row in the output.

2. Input File Format

The pipeline expects a tab-separated file with a header row containing at minimum the following columns (column names are matched case-insensitively):

Column	Type	Description
GeneID	Text	Gene identifier; multiple GeneIDs may share the same peak region
Chromosome	Text	Chromosome on which the peak resides
Peak_Start	Integer	Genomic start coordinate of the ChIP/ATAC peak
Peak_End	Integer	Genomic end coordinate of the peak
Strand	+ / -	Strand orientation of the peak
Fetched_sequence	DNA	Full genomic sequence spanning the peak region

Column	Type	Description
Matched_Sequence	DNA	Short motif sequence that was matched within the peak
Mismatches	Integer	Number of mismatches in the motif match
Position	Integer	Position of the motif match within the fetched sequence
Alignment_Score	Float	Alignment score of the motif match
Upstream	DNA	Sequence immediately upstream of the matched motif (USED FOR CLUSTERING)
Downstream	DNA	Sequence immediately downstream of the matched motif (USED FOR CLUSTERING)

Rows with missing or empty Upstream or Downstream sequences are automatically removed before clustering with a warning printed to the console.

3. Pipeline Steps

Step 1 – Data Loading & Deduplication

The TSV file is loaded and columns are resolved using case-insensitive matching. Sequences are stripped of whitespace and converted to uppercase. A unique pair_id (e.g. pair_001, pair_002, …) is assigned to each distinct (Upstream, Downstream) combination.

A gene_map dictionary is constructed that records which GeneIDs correspond to each pair_id. All downstream computation is performed on the set of unique pairs, and cluster labels are merged back to the full table at the end.

> **Why deduplicate?** In genomic datasets, genes sharing overlapping peaks will have identical flanking sequences. Clustering duplicates inflates cluster sizes and distorts distance matrices without adding information. Deduplication ensures each data point represents a biologically distinct sequence context.

Step 2 – Pairwise Alignment Distance Matrix

For each clustering mode, an N × N distance matrix is computed where N is the number of unique sequence pairs. Pairwise sequence similarity is measured using BioPython's PairwiseAligner with either:

- Global alignment (Needleman-Wunsch) – default; aligns sequences end-to-end; appropriate when sequences are expected to be globally similar
- Local alignment (Smith-Waterman) – finds the best local match; appropriate when sequences differ greatly in length or share only a conserved sub-region

The raw alignment score is normalised to a distance value in [0, 1] using the formula:

```
distance(i, j) = 1 − score(i, j) / √( score(i, i) × score(j, j) )
```

Here score(i, j) is the raw alignment score between sequences i and j, and score(i, i) is the self-alignment score of sequence i. Dividing by the geometric mean of the two self-scores normalises for sequence length and composition differences. The resulting distance is 0 for identical sequences and approaches 1 for completely dissimilar sequences.

For the combined mode, the Upstream and Downstream distance matrices are merged as a normalised weighted average:

$$\text{dist_combined} = (w_1 \times \text{dist_upstream} + w_2 \times \text{dist_downstream}) / (w_1 + w_2)$$

Default weights are $w_1 = w_2 = 0.5$ (equal contribution). Custom weights can be supplied via command-line arguments --w-upstream and --w-downstream.

Step 3 – UMAP Dimensionality Reduction

Uniform Manifold Approximation and Projection (UMAP) is applied to each N × N distance matrix to produce a 2-dimensional embedding of the sequences. UMAP operates directly on the precomputed distance matrix (metric="precomputed") and learns a low-dimensional representation that preserves both local neighbourhood structure and global topology.

Key UMAP hyperparameters:

Parameter	Description
n_neighbors	Controls the balance between local and global structure. A smaller value (e.g. 3–5) focuses on fine-grained local clusters; a larger value (e.g. 15–30) reveals broader global patterns. Automatically capped at N-1 to satisfy UMAP constraints with small datasets.
min_dist	Minimum distance between embedded points. Values close to 0 produce tightly packed, visually distinct clusters. Larger values (0.3–0.5) spread points more evenly for a global view.
random_state	Fixed seed for reproducibility (default: 42).
metric	Set to precomputed so UMAP reads the alignment-based distance matrix directly, without re-computing distances in feature space.

Why UMAP over PCA or MDS? UMAP is non-linear and better preserves cluster topology in biological sequence data compared to linear methods (PCA) or stress-minimising approaches (MDS). Unlike t-SNE, UMAP scales efficiently to larger datasets and maintains meaningful global structure between clusters.

Step 4 – K-Means Clustering on UMAP Coordinates

K-Means clustering is applied to the 2-D UMAP coordinates (not to the raw distance matrix). This two-stage approach is intentional: UMAP first projects the non-Euclidean alignment distances into a Euclidean space where K-Means can operate effectively, since K-Means assumes Euclidean geometry and requires a feature-space representation.

K-Means algorithm summary:

4. Initialise k cluster centroids (using k-means++ strategy for stable initialisation; n_init=20 restarts)
5. Assign each sequence pair to the nearest centroid in UMAP 2-D space
6. Recompute centroids as the mean of all assigned points
7. Repeat steps 2–3 until assignments no longer change or a convergence tolerance is reached

Parameter	Description
n_clusters (k)	Number of clusters to produce. Set via --n-clusters (default: 3). Should be guided by biological expectation or an elbow/silhouette analysis.
n_init	Number of K-Means runs with different initialisations (set to 20 for robust results).
random_state	Fixed seed (default: 42) for reproducibility.

Cluster labels are 1-based integers (1, 2, … k). The same label scheme is used across all three modes (Upstream, Downstream, Combined) so results can be compared directly.

Step 5 – Hierarchical Clustering (Dendrogram)

Independently of K-Means, a hierarchical (agglomerative) clustering is computed on the full N × N distance matrix. This is used exclusively to generate the dendrogram visualisation; the final cluster labels reported in the output TSV come from K-Means.

The linkage matrix Z is computed using scipy.cluster.hierarchy.linkage on the condensed (upper-triangle) form of the distance matrix. Available linkage methods:

Method	Description
average (default)	UPGMA – distance between clusters is the average of all pairwise distances between their members. Robust and widely used in phylogenetics.
ward	Minimises the total within-cluster variance at each merge step. Tends to produce compact, roughly equal-sized clusters.
complete	Uses the maximum pairwise distance between clusters (furthest-neighbour). Conservative; avoids chaining but can be sensitive to outliers.
single	Uses the minimum pairwise distance (nearest-neighbour). Can produce elongated, chained clusters.

The dendrogram is drawn with leaves coloured by their K-Means cluster assignment. A horizontal dashed red cut-line is overlaid at the height corresponding to k clusters, providing a visual bridge between the hierarchical tree structure and the K-Means partition.

4. Outputs

4.1 TSV Results File (clustering_results.tsv)

The output TSV contains all columns from the original input file plus the following appended columns:

Column	Description
kmeans_upstream	K-Means cluster label derived from Upstream sequence distances (1-based integer)
kmeans_downstream	K-Means cluster label derived from Downstream sequence distances
kmeans_combined	K-Means cluster label derived from the combined (Upstream + Downstream) distance matrix

Because clustering is performed on unique sequence pairs and labels are merged back to the full table, all GeneIDs sharing the same (Upstream, Downstream) pair will receive identical cluster labels.

4.2 Plots File (clustering_plots.png)

A 3 × 3 panel figure is generated with rows corresponding to the three clustering modes (Upstream, Downstream, Combined) and columns corresponding to the three plot types:

Panel	Clustering basis	Description
Col 0: Dendrogram	Hierarchical	Tree diagram of pairwise sequence relationships. Leaves are labelled by pair_id and GeneID(s). Leaf labels are coloured by K-Means cluster. A red dashed horizontal line marks the cut point for k clusters.
Col 1: Distance Heatmap	K-Means overlay	Symmetric N × N heatmap of pairwise alignment distances (darker = more similar). Diagonal cells are colour-filled by K-Means cluster membership to cross-reference cluster labels with raw distances.
Col 2: UMAP Scatter	K-Means	2-D scatter plot of UMAP-embedded sequence pairs. Each point represents a unique (Upstream, Downstream) pair, labelled by pair_id and coloured by K-Means cluster. Tight groupings indicate high sequence similarity.

5. Command-Line Reference

5.1 Basic Usage

```
python cluster_sequences.py <input.tsv> [OPTIONS]
```

5.2 All Arguments

Argument	Default	Description
input	(required)	Path to the tab-separated input file
--output-prefix	clustering	Prefix for output filenames (e.g. my_run produces my_run_results.tsv)
--n-clusters	3	Number of K-Means clusters k
--kmeans-seed	42	Random seed for K-Means reproducibility
--linkage	average	Linkage method for the dendrogram: single | complete | average | ward
--umap-n-neighbors	5	UMAP n_neighbors parameter (lower = more local structure)
--umap-min-dist	0.3	UMAP min_dist (smaller = tighter cluster packing in embedding)
--umap-seed	42	Random seed for UMAP reproducibility
--alignment-mode	global	Alignment algorithm: global (Needleman-Wunsch) | local (Smith-Waterman)
--w-upstream	0.5	Weight for Upstream distance matrix in combined mode
--w-downstream	0.5	Weight for Downstream distance matrix in combined mode
--no-plots	False	Flag: skip generating the plots PNG

5.3 Example Commands

Default run – 3 clusters, global alignment, average linkage:

```
python cluster_sequences.py peaks.tsv --n-clusters 3
```

5 clusters with Ward linkage and local (Smith-Waterman) alignment:

```
python cluster_sequences.py peaks.tsv --n-clusters 5 --linkage ward --alignment-mode
local
```

Tight UMAP embedding (emphasises compact clusters):

```
python cluster_sequences.py peaks.tsv --n-clusters 4 --umap-n-neighbors 3 --umap-min-
dist 0.05
```

Give Downstream sequences twice the weight in combined mode:

```
python cluster_sequences.py peaks.tsv --w-upstream 0.33 --w-downstream 0.67
```

Custom output prefix, no plots:

```
python cluster_sequences.py peaks.tsv --output-prefix my_analysis --no-plots
```

6. Software Dependencies

Install all dependencies with a single pip command:

```
pip install biopython scikit-learn scipy matplotlib seaborn pandas numpy umap-learn
```

Package	Min. version	Purpose
biopython	≥1.79	PairwiseAligner for sequence alignment (Needleman-Wunsch / Smith-Waterman)
numpy	≥1.21	Numerical arrays and matrix operations
pandas	≥1.3	TSV loading, DataFrame manipulation, result export
scipy	≥1.7	squareform distance conversion, linkage matrix, dendrogram drawing
scikit-learn	≥1.0	KMeans clustering implementation
umap-learn	≥0.5	UMAP dimensionality reduction (install as umap-learn, import as umap)
matplotlib	≥3.4	Plotting backend (Agg backend used for headless/server environments)
seaborn	≥0.11	Distance heatmap with sns.heatmap

7. Interpreting Results

7.1 Choosing k (Number of Clusters)

The number of clusters k is the most important parameter. There is no single correct value; it should be guided by:

- Biological prior knowledge – e.g. expected number of motif families or regulatory contexts
- Visual inspection of the dendrogram – count the number of major branches at a natural gap in merge heights
- Elbow method – plot within-cluster sum of squares vs k and look for the elbow point
- Silhouette analysis – higher mean silhouette scores indicate better-separated clusters

7.2 Reading the Dendrogram

In the dendrogram, the vertical axis represents merge distance (0 = identical, 1 = maximally dissimilar). Sequences joined by a low horizontal bar are highly similar. The red dashed cut-line separates the tree into k subtrees corresponding to K-Means clusters. Leaf label colours match cluster colours in the UMAP scatter plot.

7.3 Reading the UMAP Scatter

Each point in the UMAP scatter represents a unique (Upstream, Downstream) sequence pair. Points that are close together in the UMAP space had low pairwise alignment distances — i.e. their flanking sequences are similar. Well-separated islands of same-coloured points indicate robust, biologically meaningful clusters. Overlapping clusters or scattered points may indicate that k is too large or that the sequences lack strong structure.

7.4 Comparing Modes

Comparing the cluster_upstream, cluster_downstream, and cluster_combined columns in the output TSV allows you to assess whether the upstream context, downstream context, or their combination drives the grouping of genomic peaks. GeneIDs that cluster together in all three modes likely share a conserved regulatory environment, while those that only cluster in one mode may have context-specific sequence features on a single flank.

8. Pipeline Flowchart

```
Load TSV  →  Resolve columns (GeneID, Upstream, Downstream)
Deduplicate on unique (Upstream, Downstream) pairs  →  assign pair_id
For each mode  (Upstream / Downstream / Combined):
    ├─ Compute N×N alignment distance matrix  (Needleman-Wunsch or Smith-Waterman)
    ├─ Run UMAP  (precomputed metric)  →  2-D coordinates
    ├─ Run K-Means on UMAP coordinates  →  cluster labels (1-based)
    └─ Compute hierarchical linkage  →  linkage matrix Z  (for dendrogram)
Merge cluster labels back to all GeneID rows  →  save TSV
Generate 3×3 plot grid:  Dendrogram  |  Distance Heatmap  |  UMAP Scatter
```